Alibaba Cloud

Large design

Architecture proposal

Date	Revisions	Products	Catalog checked
31 October 2026	2	28	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Summary

The design

Let's architect your cross-border e-commerce platform for Southeast Asia, building from users inward. First, the edge layer to handle global acceleration, DDoS protection, and caching product images close to users. Now the core region with a VPC, highly available load balancers, and auto-scaling compute for 10x flash sales. For data, we need cloud-native relational DB, Redis cache for flash sales, object storage for images, and search. Adding the AI shopping assistant chatbot using Model Studio for Qwen LLM and RocketMQ for async order events. Finally, observability, security, and network egress to complete the highly available architecture.

28 products, 25 connections, 2 revisions

Pinned requirements

No requirements were pinned.

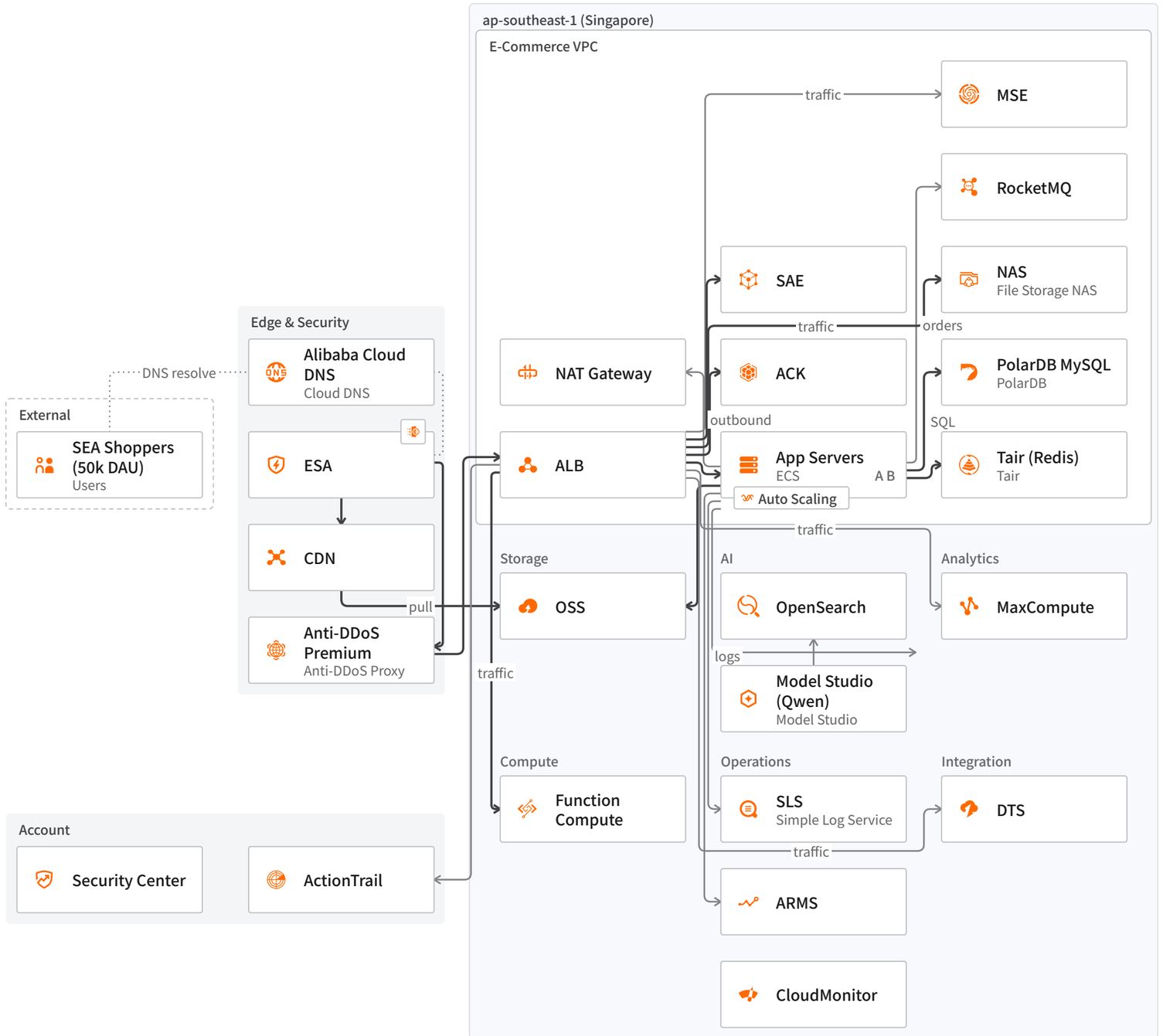
Constraints

No region or product constraints were set.

Assumptions

No assumptions were recorded.

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

SEA Shoppers (50k DAU)
External

Role in this design
End users browsing and buying across Southeast Asia

What it does
People reaching the system over the internet from browsers or apps.

Official documentation
Documentation link not yet available

Security Center

Account · Security

Role in this design
Host threat protection and compliance posture

What it does
Workload protection: vulnerabilities, baselines, threat detection and Agentic SOC (SIEM/SOAR).

Official documentation
www.alibabacloud.com/product/security_center

ActionTrail

Account · Migration & O&M Management

Role in this design
Added to make the diagram bigger: Audit trail of console and API actions, delivered to SLS or OSS.

What it does
Audit trail of console and API actions, delivered to SLS or OSS.

Official documentation
www.alibabacloud.com/product/actiontrail

Alibaba Cloud DNS

Alibaba Cloud DNS
Edge & Security · Domain Names and Websites

Role in this design
Low-latency authoritative DNS routing for SEA shoppers

What it does
Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation
www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge & Security · Networking & CDN

Role in this design
Dynamic acceleration and edge security for cross-border latency

What it does
Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation
www.alibabacloud.com/product/esa

Web Application Firewall (WAF)

Edge & Security · On ESA

Role in this design

L7 protection against bots and OWASP attacks

What it does

WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation

www.alibabacloud.com/product/web-application-firewall

CDN

Edge & Security · Networking & CDN

Role in this design

Cache heavy product images at SEA edge nodes

What it does

Caches static content on edge nodes close to users.

Official documentation

www.alibabacloud.com/product/content-delivery-network

Anti-DDoS Pro/Premium (Anti-DDoS Proxy)

Anti-DDoS Premium

Edge & Security · Security

Role in this design

Volumetric attack mitigation for flash sale traffic spikes

What it does

Proxy scrubbing of large L3-L7 DDoS attacks via CNAME or IP; Premium for overseas origins.

Official documentation

www.alibabacloud.com/product/ddos

NAT Gateway

ap-southeast-1 (Singapore) · E-Commerce VPC · Networking & CDN

Role in this design

Secure outbound internet for ECS patching and APIs

What it does

Gives private VPC resources outbound internet (SNAT) and mapped inbound ports (DNAT).

Official documentation

www.alibabacloud.com/product/nat

Application Load Balancer (ALB)

ap-southeast-1 (Singapore) · E-Commerce VPC · Networking & CDN

Role in this design

L7 load balancing across multi-zone ECS fleet

What it does

Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation

www.alibabacloud.com/product/server-load-balancer

Serverless App Engine (SAE)

ap-southeast-1 (Singapore) · E-Commerce VPC · Computing

Role in this design

Added to make the diagram bigger: Serverless PaaS for containers and Java apps, without servers or Kubernetes to run.

What it does

Serverless PaaS for containers and Java apps, without servers or Kubernetes to run.

Official documentation

www.alibabacloud.com/product/severless-application-engine

Container Service for Kubernetes (ACK)

ap-southeast-1 (Singapore) · E-Commerce VPC · Container

Role in this design

Added to make the diagram bigger: Managed Kubernetes (ACK Pro) with node pools, CSI storage, Ingress and an AI suite.

What it does

Managed Kubernetes (ACK Pro) with node pools, CSI storage, Ingress and an AI suite.

Official documentation

www.alibabacloud.com/product/kubernetes

Elastic Compute Service (ECS)

App Servers

ap-southeast-1 (Singapore) · E-Commerce VPC · Zones A, B

Role in this design

Stateless web/app tier handling business logic

What it does

Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation

www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

ap-southeast-1 (Singapore) · E-Commerce VPC · On App Servers

Role in this design

Scale ECS fleet 10x during flash sales automatically

What it does

Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation

www.alibabacloud.com/product/auto-scaling

Microservices Engine (MSE)

ap-southeast-1 (Singapore) · E-Commerce VPC · Middleware

Role in this design

Added to make the diagram bigger: Managed Nacos or ZooKeeper registry and config, plus microservice governance.

What it does

Managed Nacos or ZooKeeper registry and config, plus microservice governance.

Official documentation

www.alibabacloud.com/product/microservices-engine

ApsaraMQ for RocketMQ

ap-southeast-1 (Singapore) · E-Commerce VPC · Middleware

Role in this design
Decouple order processing and handle flash sale spikes

What it does
Managed RocketMQ 5.x for transactional, scheduled and ordered messaging.

Official documentation
www.alibabacloud.com/product/mq

File Storage NAS

ap-southeast-1 (Singapore) · E-Commerce VPC · Storage

Role in this design
Added to make the diagram bigger: Managed NFS/SMB shared file system mounted by ECS, ACK or FC.

What it does
Managed NFS/SMB shared file system mounted by ECS, ACK or FC.

Official documentation
www.alibabacloud.com/product/nas

PolarDB

PolarDB MySQL
ap-southeast-1 (Singapore) · E-Commerce VPC · Database

Role in this design
Cloud-native HA database for orders and inventory

What it does
Cloud-native relational database (MySQL, PostgreSQL, Oracle-compatible) with compute-storage separation.

Official documentation
www.alibabacloud.com/product/polardb

Tair (Redis® OSS-Compatible)

ap-southeast-1 (Singapore) · E-Commerce VPC · Database

Role in this design
In-memory cache for cart, sessions, flash sale stock

What it does
Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation
www.alibabacloud.com/product/tair

Object Storage Service (OSS)

ap-southeast-1 (Singapore) · regional services · Storage

Role in this design
Store product images served via CDN

What it does
Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation
www.alibabacloud.com/product/object-storage-service

Function Compute (FC)

ap-southeast-1 (Singapore) · regional services · Computing

Role in this design

Added to make the diagram bigger: Event-driven serverless functions and web functions, including GPU model serving.

What it does

Event-driven serverless functions and web functions, including GPU model serving.

Official documentation

www.alibabacloud.com/product/function-compute

OpenSearch

ap-southeast-1 (Singapore) · regional services · AI & Machine Learning

Role in this design

Fast product search with LLM conversational capabilities

What it does

Managed search, including vector search and LLM-based conversational search (RAG) editions.

Official documentation

www.alibabacloud.com/product/opensearch

Alibaba Cloud Model Studio

Model Studio (Qwen)
ap-southeast-1 (Singapore) · regional services · AI & Machine Learning

Role in this design

AI shopping assistant chatbot with RAG over products

What it does

Managed model APIs for Qwen (such as qwen3.8-max) and third-party models; OpenAI-compatible.

Price

Pay per token; qwen3.8-max $2 per 1M input and $6 per 1M output tokens (Singapore); 1M free tokens per model for 90 days, as of 2026-10-31

Official documentation

www.alibabacloud.com/product/modelstudio

Simple Log Service (SLS)

ap-southeast-1 (Singapore) · regional services · Storage

Role in this design

Centralized log collection and alerting

What it does

Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation

www.alibabacloud.com/product/log-service

Application Real-Time Monitoring Service (ARMS)

ap-southeast-1 (Singapore) · regional services · Middleware

Role in this design

APM tracing for microservices performance monitoring

What it does

APM, browser and synthetic monitoring, tracing and alerting for applications.

Official documentation

www.alibabacloud.com/product/arms

CloudMonitor

ap-southeast-1 (Singapore) · regional services · Migration & O&M Management

Role in this design

Infrastructure metrics and auto-scaling triggers

What it does

Resource metrics, alarms and site monitoring; CMS 2.0 adds APM and AI observability.

Official documentation

www.alibabacloud.com/product/cloud-monitor

MaxCompute

ap-southeast-1 (Singapore) · regional services · Analytics Computing

Role in this design

Added to make the diagram bigger: Serverless exabyte-scale data warehouse for batch SQL and Spark.

What it does

Serverless exabyte-scale data warehouse for batch SQL and Spark.

Official documentation

www.alibabacloud.com/product/maxcompute

Data Transmission Service (DTS)

ap-southeast-1 (Singapore) · regional services · Database

Role in this design

Added to make the diagram bigger: Database migration, real-time synchronization and change data capture.

What it does

Database migration, real-time synchronization and change data capture.

Official documentation

www.alibabacloud.com/product/data-transmission-service

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
SEA Shoppers (50k DAU)	Alibaba Cloud DNS	DNS resolve	DNS resolution
Alibaba Cloud DNS	ESA		DNS resolution
ESA	Anti-DDoS Premium		Data
ESA	CDN	images	Data
Anti-DDoS Premium	ALB	HTTPS	Data
ALB	App Servers		Data
App Servers	PolarDB MySQL	SQL	Data
App Servers	Tair (Redis)	cache	Data
App Servers	OSS	upload	Data
CDN	OSS	pull	Data
App Servers	OpenSearch	query	Data
App Servers	Model Studio (Qwen)	chat API	Data
Model Studio (Qwen)	OpenSearch	RAG	Data
App Servers	RocketMQ	orders	Data
App Servers	SLS	logs	Data
App Servers	ARMS	traces	Data
App Servers	NAT Gateway	outbound	Data
ALB	ACK	traffic	Data
ALB	MSE	traffic	Data
ALB	SAE	traffic	Data
ALB	Function Compute	traffic	Data
ALB	DTS	traffic	Data
ALB	MaxCompute	traffic	Data
ALB	NAS	traffic	Data

From	To	Carries	Type
ALB	ActionTrail	traffic	Data

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 35.8 s

Ask

We are launching a cross-border e-commerce site for Southeast Asia: about 50k daily active users, 10x traffic spikes during flash sales, product search, lots of product images, and we want an AI shopping-assistant chatbot. It must be highly available.

Changes

+ Users · SEA Shoppers (50k DAU) · in External
+ Cloud DNS · Alibaba Cloud DNS · in Edge & Security
+ Anti-DDoS Proxy · Anti-DDoS Premium · in Edge & Security
+ WAF · on ESA
+ ESA · in Edge & Security
+ CDN · in Edge & Security
+ 20 connections
+ ALB · in E-Commerce VPC
+ ECS · App Servers (AZ-A)
+ Auto Scaling · on ECS
+ PolarDB · PolarDB MySQL · in E-Commerce VPC
+ Tair (Redis) · in E-Commerce VPC
+ OSS · in E-Commerce VPC
+ OpenSearch · in E-Commerce VPC
+ Model Studio · Model Studio (Qwen) · in E-Commerce VPC
+ RocketMQ · in E-Commerce VPC
+ SLS · in E-Commerce VPC
+ ARMS · in E-Commerce VPC
+ CloudMonitor · in E-Commerce VPC
+ Security Center · in E-Commerce VPC
+ NAT Gateway · in E-Commerce VPC

Revision 2

Drawn in 3.5 s

Ask

oh i also want waf

Changes

No change to the diagram.